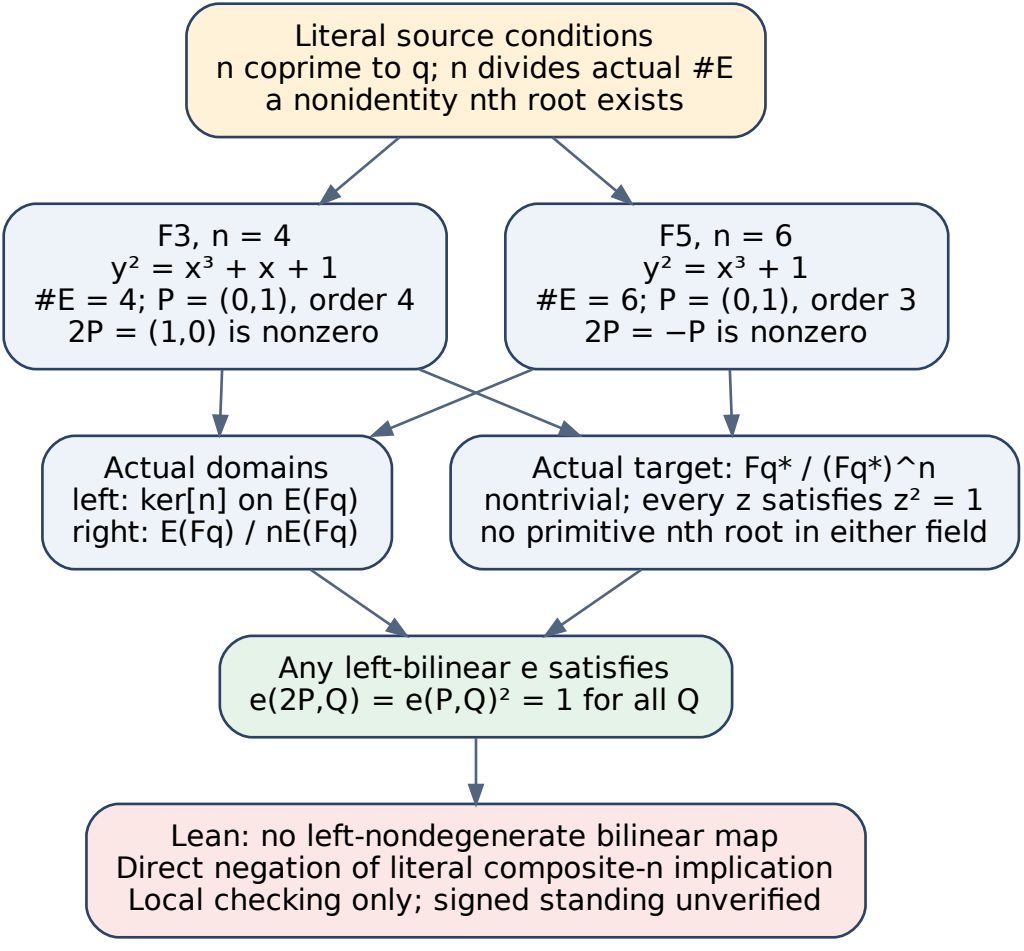
Literal source conditions
n coprime to q; n divides actual #E
a nonidentity nth root exists
F3, n = 4
y² = x³ + x + 1
#E = 4; P = (0,1), order 4
2P = (1,0) is nonzero
F5, n = 6
y² = x³ + 1
#E = 6; P = (0,1), order 3
2P = −P is nonzero
Actual domains
left: ker[n] on E(Fq)
right: E(Fq) / nE(Fq)
Actual target: Fq* / (Fq*)^n
nontrivial; every z satisfies z² = 1
no primitive nth root in either field
Any left-bilinear e satisfies
e(2P,Q) = e(P,Q)² = 1 for all Q
Lean: no left-nondegenerate bilinear map
Direct negation of literal composite-n implication
Local checking only; signed standing unverified